

CSA1715-ARTIFICIAL INTELLIGENCE

LAB PROGRAM- 17

18-08-2026

17. PROLOG PROGRAM TO SUM THE INTEGERS FROM 1 TO N

AIM

To write and execute a Prolog program to find the sum of all integers from 1 to a given positive integer N using recursion. The program demonstrates the use of recursive predicates and arithmetic operations in Prolog.

ALGORITHM

1. Start the program.
2. Define a predicate `sum_to_n(N, Sum)` to calculate the sum of integers from 1 to N.
3. Define the base case: when N is 0, the sum is 0.
4. For a positive value of N, subtract 1 from N and recursively find the sum up to N-1.
5. Add N to the recursively obtained sum.
6. Display the final sum.
7. Stop the program.

PROLOG CODE

% Program to find sum of integers from 1 to N

`sum_to_n(0, 0).`

`sum_to_n(N, Sum) :-`

`N > 0,`

`N1 is N - 1,`

sum_to_n(N1, Sum1),
Sum is N + Sum1.

SAMPLE INPUT

?- sum_to_n(5, S).

SAMPLE OUTPUT

S = 15.

RESULT

Thus, the Prolog program to calculate the sum of integers from 1 to N was successfully implemented and executed. For the input value 5, the calculated sum is 15.